

Uncertainty Estimation with Limited Data through Graph-Based Information Sharing

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Abstract—Prediction intervals give a range of plausible values for a future observation. Estimating these intervals is challenging when a location has limited recent data. We propose a variance-weighted graph-smoothing method that shares scale information between connected locations, followed by calibration to adjust interval widths. We evaluate two applications, traffic-speed forecasting and COVID-19 hospital-admission forecasting. In each application, we consider a target location with no history, limited history, or its full available history for interval estimation. Other locations retain their available data. We evaluate intervals using the Winkler score, which favors narrow intervals and penalizes observations outside them. Across the three history settings and both applications, graph smoothing achieves lower or comparable mean scores than the comparison methods. These results show how graph-based information sharing supports prediction intervals across different levels of local data availability.

Index Terms—Machine learning, conformal prediction, graph smoothing, spatio-temporal data, uncertainty quantification, traffic forecasting, epidemic forecasting.

I. INTRODUCTION

A point forecast gives one estimate of a future observation, while a prediction interval describes how far the observation may differ from that estimate. Traffic observations form graph-structured time series, with a speed sequence at each sensor. Graph-based models predict speeds by capturing spatio-temporal dependencies across sensors and time [1], [2]. These predictions are uncertain because traffic conditions vary and models learn from limited data [3], [4].

Estimating a prediction interval becomes difficult when a target location has limited recent data for interval estimation. In traffic networks, this can arise after a sensor is activated or when its earlier observations are unavailable. We use cold start to describe having no past data available for interval estimation. Changing conditions can also make recent observations more relevant than older ones [5]. A scale estimated from a short period can be sensitive to which observations are available. Connected locations can provide additional information [6], but forecast errors, the differences between observed and predicted values, can have different magnitudes across locations. Sharing information therefore requires a balance between the target location’s data and data from its neighbors.

We propose a post-hoc method that uses both graph connections and the estimated reliability of location-specific scales

to construct prediction intervals. Estimates with lower variance receive greater weight during graph smoothing, allowing connected locations to inform a target scale estimated from limited data. A separate split-conformal quantile rule calibrates interval widths [7]. We evaluate how Winkler scores change as the target location’s available data decreases.

Our main evaluation uses forecasts based on a Spatio-Temporal Graph Convolutional Network (STGCN) [1] on the METR-LA and PEMS-BAY datasets [2]. With 12, 24, or 48 recent target slots, Graph Smoothing reduces mean Winkler score [8] by 3.3–6.6% relative to pooling in Los Angeles and by 6.7–16.4% relative to separate sensor-specific scales in Bay Area. A second application uses COVIDcast hospital-admission data across U.S. states [9], where graph smoothing lowers mean scaled Winkler scores relative to pooling when target data are limited.

The remainder of this paper is organized as follows. Section II reviews related work, followed by the proposed method in Section III. Section IV describes the experimental setup for both applications, while Section V presents and discusses their results. Section VI concludes the paper.

II. RELATED WORK

We review related approaches to prediction intervals for time series and graph data. Split conformal prediction fits a model on one part of the data and uses another part to set interval widths. Keeping calibration separate from model fitting supports its marginal coverage guarantee under exchangeability [7]. Lei et al. [7] also study a locally weighted extension that scales absolute prediction errors before calibration, allowing interval widths to vary across inputs. Romano et al. [10] develop CQR, which calibrates lower and upper quantile predictions. Both approaches adapt interval widths to differences across inputs.

Time-series methods address dependence and distribution shift through sequential updates. Gibbs and Candès [5] introduce ACI, which adjusts the calibration level according to whether recent intervals contain the observations. Xu and Xie [11] develop EnbPI, which combines bootstrap ensemble predictions with an updated collection of prediction errors. Their analysis bounds coverage gaps under assumptions on

temporal dependence and estimation accuracy. These methods use new observations to update intervals over time.

Graph-based methods use relationships between nodes when constructing prediction sets. Lunde et al. [12] establish conformal validity for regression with network-derived features under joint exchangeability assumptions. Baghershahi et al. [13] propose GraphLCP, which uses graph connections and node features to weight calibration samples through a Personalized PageRank kernel. For correlated time series, Cini et al. [6] introduce CoRel, which learns relationships between time series and uses an STGNN to estimate prediction-error quantiles. Guo et al. [14] propose SCALE, which combines graph-wavelet decomposition with a learned quantile model and evaluates it on traffic datasets.

Conformal prediction has also been used to estimate uncertainty in epidemic forecasts. Wohlfender et al. [15] apply ACI to construct prediction intervals for machine learning forecasts of COVID-19 hospital admissions. For evaluating epidemic forecasts, Bracher et al. [16] describe interval scores that account for both interval width and observations outside the intervals. These works address interval construction and evaluation in the hospital-admissions application considered here.

Our contribution is a variance-weighted graph smoother for interval estimation when a target location has little or no recent data. Unlike approaches that weight calibration samples [13], we regularize location-specific log-scales using their estimated variances. Prior work learns uncertainty models [6], [14], while our scale estimator is obtained by solving a graph-regularized linear system. This provides a post-hoc approach to sharing scale information across locations before calibration.

III. GRAPH-SMOOTHED PREDICTION INTERVALS

Let $V = \{1, \dots, p\}$ be the nodes of a graph, with observations y_{jt} and forecasts $\hat{y}_{jt}$ at node $j \in V$ and time t . For target node $i \in V$, radius r_i defines

$$I_{it} = [\hat{y}_{it} - r_i, \hat{y}_{it} + r_i]. \quad (1)$$

For a chosen miscoverage level $\alpha \in (0, 1)$, we target coverage $1 - \alpha$ separately for each forecast horizon and amount of target-node data.

A. Starting Scales

For usable estimation times $\mathcal{E}_j$ at node j and count $n_j = |\mathcal{E}_j|$, define scaled absolute forecast errors

$$d_{jt} = \frac{|y_{jt} - \hat{y}_{jt}|}{\tau_j}, \quad \tau_j > 0. \quad (2)$$

The normalization factor τ_j is fixed before interval estimation and remains available at cold start.

For sorted values $x_{(1)} \leq \dots \leq x_{(n)}$, define $Q_{1-\alpha}(x) = x_{(1 + \lceil (1-\alpha)(n-1) \rceil)}$. With floor $\epsilon = 10^{-6}$, the pooled fallback and starting scales are

$$\begin{aligned} s_{-i} &= \max\{\epsilon, Q_{1-\alpha}(\{d_{jt} : j \neq i, t \in \mathcal{E}_j\})\}, \\ s_j &= \begin{cases} \max\{\epsilon, Q_{1-\alpha}(\{d_{jt} : t \in \mathcal{E}_j\})\}, & n_j > 0, \\ s_{-i}, & n_j = 0. \end{cases} \end{aligned} \quad (3)$$

From an older period of length T , sample B consecutive windows of length $\ell = \min(m, \lfloor T/2 \rfloor)$ for reference length $m \leq T$, with starts drawn uniformly with replacement. Let $\mathcal{E}_{jb}$ contain the usable times for node j in window b , with $n_{jb} = |\mathcal{E}_{jb}|$. For $n_{jb} > 0$, define

$$u_{jb} = \log[\max\{\epsilon, Q_{1-\alpha}(\{d_{jt} : t \in \mathcal{E}_{jb}\})\}].$$

Using these nonempty windows, we estimate

$$\begin{aligned} \hat{v}_j &= \frac{\ell}{m} \max\{10^{-8}, \text{Var}_b(u_{jb})\}, \\ c_j &= n_j^{\text{ref}} \hat{v}_j, \quad w_j = \frac{n_j}{c_j}. \end{aligned} \quad (4)$$

Here Var_b is the sample variance across usable windows. For $m < T$, $n_j^{\text{ref}} = \text{round}[(m/\ell) \text{median}_b n_{jb}]$. For $m = T$, it counts all usable observations in the older period. Dependent-quantile asymptotics motivate the approximation $\text{Var}(\log \hat{s}_j) \approx c_j/n_j$ and inverse-variance weights [17]. A limited-data target uses $\text{median}_{j \neq i} c_j$ over finite positive coefficients in place of c_i . We set $w_j = 0$ when $n_j = 0$ or variance is unavailable.

B. Graph Smoothing

Let $A = (a_{jk})$ be the symmetric weighted adjacency matrix, $E = \{\{j, k\} : j < k, a_{jk} > 0\}$, and $\mathcal{O} = \{j \in V : w_j > 0\}$. For $\lambda \geq 0$, weighted graph Tikhonov regularization [18] gives

$$\begin{aligned} \hat{\mathbf{z}} \in \arg \min_{\mathbf{z}} \left\{ \sum_{j \in \mathcal{O}} w_j (z_j - \log s_j)^2 \right. \\ \left. + \lambda \sum_{\{j,k\} \in E} a_{jk} (z_j - z_k)^2 \right\}. \end{aligned} \quad (5)$$

Weighted least squares gives more influence to estimates with lower estimated variance, while Laplacian regularization penalizes differences between connected log-scales. The convex quadratic objective yields a linear system, with $W = \text{diag}(w_j)$ and graph Laplacian $L = \text{diag}(A1) - A$,

$$(W + \lambda L)\hat{\mathbf{z}} = W \log \mathbf{s}, \quad \tilde{s}_j = e^{\hat{z}_j} > 0. \quad (6)$$

Log-space smoothing penalizes relative differences through $\hat{z}_j - \hat{z}_k = \log(\tilde{s}_j/\tilde{s}_k)$.

For $\lambda > 0$, each connected component intersecting $\mathcal{O}$ has a unique solution to (6). Within these components, nodes outside $\mathcal{O}$ receive a harmonic extension, a weighted average of neighboring log-scales. Components disjoint from $\mathcal{O}$ use s_{-i} . At $\lambda = 0$, we define $\tilde{s}_j = s_j$ for $j \in \mathcal{O}$ and adopt the fallback convention $\tilde{s}_j = s_{-i}$ otherwise.

The degrees-of-freedom ratio is the normalized trace of the linear smoother,

$$\rho(\lambda) = \frac{\text{tr}[(W + \lambda L)^{-1}W]}{|\mathcal{O}|}, \quad \lambda > 0. \quad (7)$$

Matrices are restricted to components intersecting $\mathcal{O}$, with $\rho(0) = 1$. Smaller ratios give stronger sharing. Tuning selects one ratio, and each target case's weights determine λ . The strongest-smoothing grid setting uses a large finite λ to approximate a common scale within each connected component.

C. Calibration and Comparisons

Let $\mathcal{C}$ index usable node–time calibration pairs and $\mathcal{C}_i = \{(j, t) \in \mathcal{C} : j \neq i\}$. For N sorted scores, the split-conformal rule [7] gives $Q_{\text{cal}}(x) = x_{(\lceil (1-\alpha)(N+1) \rceil)}$, with $x_{(N+1)} = +\infty$.

Set $b_j = 1$ for Global, $b_j = s_j$ for Local, and $b_j = \tilde{s}_j$ for Graph Smoothing. The nonconformity scores d_{jt}/b_j measure scale-adjusted forecast errors. Each method calculates

$$\begin{aligned} q_i &= Q_{\text{cal}}(\{d_{jt}/b_j : (j, t) \in \mathcal{C}_i\}), \\ r_i &= \tau_i q_i b_i. \end{aligned} \quad (8)$$

The multiplier q_i adjusts interval size toward coverage $1 - \alpha$. Local provides an unsmoothed reference under the same calibration procedure. We use the split-conformal quantile rule without assuming exchangeability across nodes and time. We therefore report empirical coverage on test data.

IV. EXPERIMENTAL SETUP

We evaluate whether Graph Smoothing improves prediction intervals with limited data. We vary the history available for interval estimation at one target node while all other nodes retain their full available history. We repeat this experiment with every node as the target in each dataset. We compare Global, Local, and Graph Smoothing across target histories and forecast horizons, with $\alpha = 0.10$.

We split data chronologically into training (60%), validation (10%), tuning (10%), calibration (10%), and testing (10%). We use application-specific tuning windows to accommodate the sampling frequency and forecast horizon. Each reference variance estimate uses 200 sampled windows. For each target history and forecast horizon, we average tuning Winkler scores equally across locations and select the least smoothing among 15 settings within 1% of the lowest mean score. Interval radii remain fixed during testing.

A. Traffic Forecasting

We use METR-LA from Los Angeles and PEMS-BAY from the San Francisco Bay Area [2], which contain traffic speeds in mph at five-minute intervals (Table I). The DCRNN graphs connect sensor nodes with weights that decrease with road-network distance [2]. For Graph Smoothing, we symmetrize each graph using the larger of the two directed weights and remove self-connections. Fig. 1 shows the Los Angeles network.

TABLE I
TRAFFIC DATASETS WITH FIVE-MINUTE OBSERVATIONS. EDGES ARE UNDIRECTED AND EXCLUDE SELF-CONNECTIONS.

Dataset	Sensors	Time points	Missing (%)	Edges
Los Angeles	207	34,272	8.109	1,313
Bay Area	325	52,116	0.003	2,079

We train an STGCN-based predictor [1] separately for each dataset. It uses 12 past speed measurements to predict the next 12 steps. We evaluate forecasts 15, 30, and 60 minutes ahead. Training uses Adam with a learning rate of 10^{-3} and

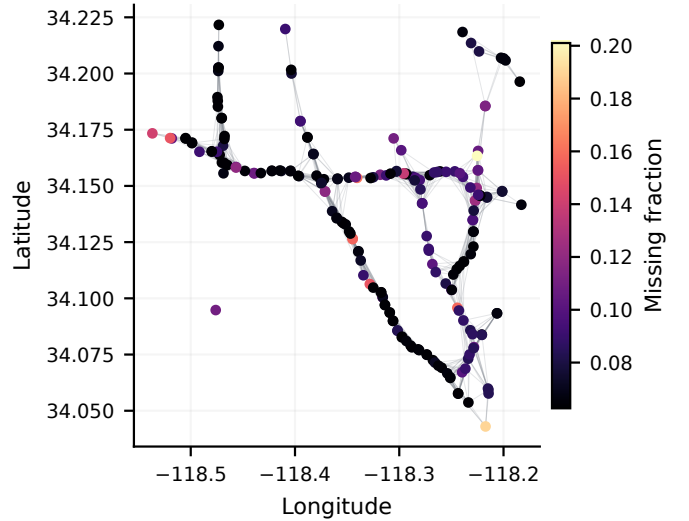


Fig. 1. Los Angeles sensor network. Node colors show the fraction of readings treated as missing.

a batch size of 32. We select the checkpoint with the lowest validation MAE. We treat nonfinite and nonpositive speeds as missing. Forward-fill imputation replaces missing inputs with the sensor’s last available speed [19], with its training-period median as the fallback when no earlier reading is available.

Tuning uses an approximate 60/20/20 split for older scale estimates, calibration and recent scales, and candidate scoring, with 12-slot boundary offsets. Among observations available before scoring, the middle portion’s latest 48 slots estimate recent scales and earlier slots calibrate intervals. With $\tau_j = 1$, each target sensor uses 0, 12, 24, or 48 recent five-minute slots, or its full available data. Other sensors retain their full available data.

For testing, full data combines tuning data with the latest 48 calibration slots available before testing begins. Limited-data targets use their assigned recent slots. Earlier calibration data from other sensors determines interval multipliers. Evaluation requires an observed target speed and 12 observed input measurements at the target sensor.

B. COVID-19 Hospital Admissions

We use daily confirmed COVID-19 hospital admissions from HHS through COVIDcast [9], distributed with the geographic graph at <https://github.com/Alistair-Turcan/EpiPatch>. The data span 978 days from August 23, 2021, to April 26, 2024, with $y_{jt} \geq 0$ and no missing values. The largest connected component covers the 48 contiguous states and Washington, D.C., with $|V| = 49$, $|E| = 109$, and $a_{jk} \in \{0, 1\}$.

Each node represents a state. For seven-day-ahead forecasts, we use

$$\begin{aligned} \tau_j &= \sigma_{j, \text{train}}, \\ I_{it} &= [\max(0, \hat{y}_{it} - r_i), \hat{y}_{it} + r_i], \end{aligned} \quad (9)$$

where $\sigma_{j, \text{train}}$ is the state’s training-period standard deviation. Each state serves as target i with history $k \in \{0, 7, 14, \text{full}\}$.

Let $\mathcal{T}$ contain the 92 tuning dates and $\mathcal{R}_k$ the latest k available calibration dates, with $\mathcal{R}_0 = \emptyset$. The estimation sets are

$$\mathcal{E}_j = \begin{cases} \mathcal{R}_k, & j = i, k \in \{0, 7, 14\}, \\ \mathcal{T} \cup \mathcal{R}_{14}, & \text{otherwise.} \end{cases} \quad (10)$$

Tuning uses the first 32 dates for older scale estimates and the final 21 for scoring. Among intervening observations available before scoring, the latest 14 dates estimate recent scales and earlier dates calibrate intervals. For testing, the earlier 78 calibration dates from states $j \neq i$ determine q_i . Testing covers 92 daily forecasts per state from January 26 to April 26, 2024.

C. Evaluation Metrics

We report Winkler score, empirical coverage, and interval width. Winkler score penalizes wide intervals and missed observations, with lower scores preferred [8], [16]. Coverage is the fraction of test observations inside the intervals. Traffic scores and widths use mph. Hospital scores are divided by each state’s training-period standard deviation τ_j , while widths use admissions. We average metrics within each location and then equally across locations, separately for each forecast horizon and target history.

Code and instructions for reproducing the experiments are available at <https://github.com/Arashlf/graph-smoothed-intervals-mldb>.

V. RESULTS AND DISCUSSION

We examine how the three approaches compare as the amount of available data changes.

A. Traffic Forecasting Results

Table II compares scores, coverage, and widths for forecasts 30 minutes ahead using the target sensor’s latest 24 five-minute slots. Fig. 2 shows the broader comparison in Los Angeles. When we use data from the target sensor’s latest 12, 24, or 48 five-minute slots, Graph Smoothing lowers mean Winkler score by 3.3–6.6% relative to Global and 14.9–22.1% relative to Local. These ranges cover the three amounts of available data and all forecast horizons. When we use no data from the target sensor to estimate its interval, Graph Smoothing lowers the score by 3.5–7.0% relative to Global.

TABLE II
TEST RESULTS FOR FORECASTS 30 MINUTES AHEAD USING THE TARGET SENSOR’S LATEST 24 FIVE-MINUTE SLOTS. VALUES ARE SENSOR AVERAGES, WITH A 90% COVERAGE TARGET. BOLD MARKS THE LOWEST MEAN WINKLER SCORE IN EACH DATASET.

Method	Winkler score (mph)	Coverage (%)	Width (mph)
<i>Los Angeles</i>			
Global	36.19	90.14	16.50
Local	43.93	80.53	15.94
Graph Smoothing	34.66	89.82	15.88
<i>Bay Area</i>			
Global	21.64	88.27	7.06
Local	23.59	78.64	6.42
Graph Smoothing	21.51	87.17	6.77

In Bay Area, Graph Smoothing lowers mean Winkler score by 6.7–16.4% relative to Local when we use data from the target sensor’s latest 12, 24, or 48 five-minute slots. Its scores are close to Global, differing by less than 1.3%. When we use no data from the target sensor to estimate its interval, Graph Smoothing gives the same intervals as Local.

Fig. 3 shows how the score differences vary across Los Angeles sensors. In the cases using 0, 12, 24, or 48 five-minute slots for the target sensor, Graph Smoothing gives lower scores than Global for about 75–80% of sensors. In Bay Area, it gives lower scores for about 50–58% of sensors in the cases using 12, 24, or 48 slots, and the median reductions remain near zero. These results show that the average score reductions are shared by more sensors in Los Angeles than in Bay Area.

When we use the target sensor’s full available data, Graph Smoothing gives the same intervals as Local for all three Bay Area forecast horizons and for 15- and 30-minute Los Angeles forecasts. For 60-minute Los Angeles forecasts, their mean Winkler scores differ by about 0.1%, with Local slightly lower. At zero target history, Local uses a pooled starting scale from other sensors while retaining scale-normalized calibration. Its mean scores are close to Global’s, with slightly different intervals. With a short history, Local replaces this pooled fallback with a scale estimated from only a few target observations. That estimate can be unstable, which helps explain why using a short history can perform worse than using the pooled fallback.

Local substantially undercovers with limited target data. For 30-minute forecasts using 24 recent slots, its mean coverage is 80.5% in Los Angeles and 78.6% in Bay Area, roughly 10 percentage points below the 90% target. Graph Smoothing raises coverage to 89.8% and 87.2%, respectively, bringing both substantially closer to the target.

Fig. 4 shows the intervals for one Los Angeles sensor. Local produces a narrow interval that misses many observed speeds, while Global produces a much wider interval. The Graph Smoothing interval is wider than Local and narrower than Global. We selected this sensor as an example, while the results above summarize all sensors.

B. COVID-19 Hospital Admissions Results

Graph Smoothing also lowers mean scaled Winkler scores in the hospital-admissions application. Fig. 5 compares the three approaches as the target state’s available data changes. With 0, 7, and 14 days of target data, Graph Smoothing reduces mean scaled Winkler score by 10.2%, 5.6%, and 3.1% relative to Global, respectively. It also gives lower mean scores than Local in each of these cases. With full available data, Graph Smoothing and Local give the same intervals.

With limited target data, Graph Smoothing gives mean interval widths of 60.5–64.0 admissions, compared with 74.5 for Global. Although narrower than Global’s, these intervals have mean empirical coverage of 96.7–98.7%, above the nominal 90% target. This application demonstrates the use of the same scale-sharing method around hospital-admission forecasts.

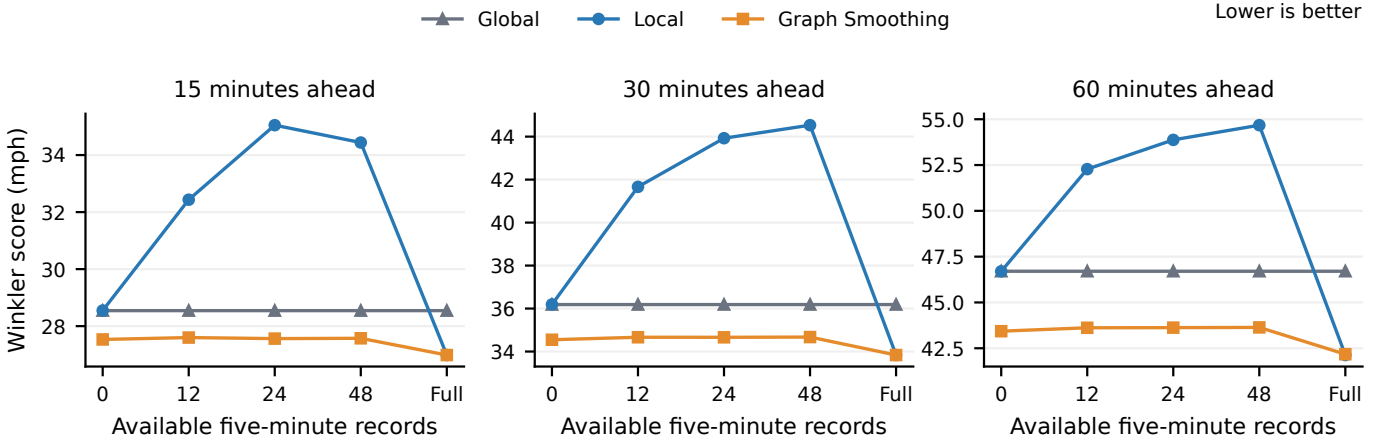


Fig. 2. Mean Winkler score by target history in Los Angeles. Full uses all available data for interval estimation, where Local and Graph Smoothing nearly overlap.

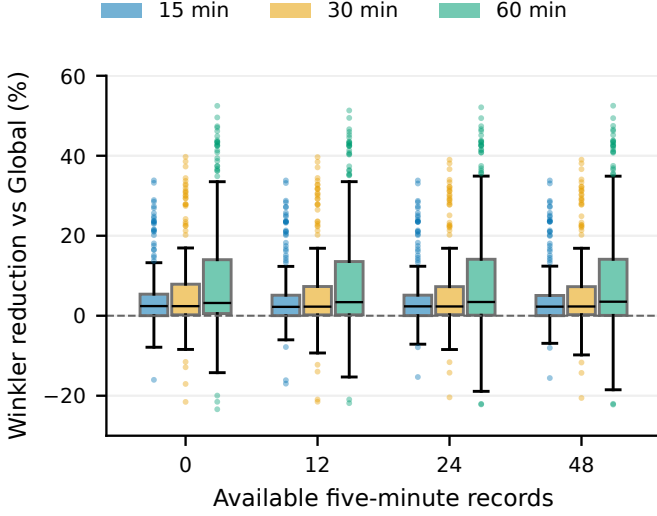


Fig. 3. Sensor-level percentage reductions in Winkler score relative to Global in Los Angeles. Positive values favor Graph Smoothing. Boxes span the interquartile range (IQR) of sensor-level reductions, with a line at the median.

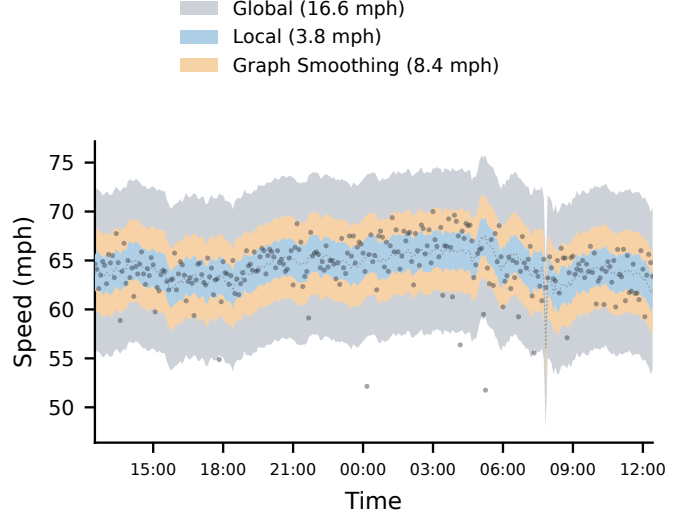


Fig. 4. Illustrative intervals for a Los Angeles sensor with 12 recent five-minute slots and 30-minute-ahead forecasts. Bands show prediction intervals, and dots show observed speeds.

C. Discussion and Limitations

Graph-based scale sharing offers a practical approach to interval estimation with limited local data. Its gains vary across datasets, and our evaluation covers one test period per dataset. The results address scarcity at one target at a time, with full available histories at other nodes. The comparisons assess the complete method, leaving the separate contributions of variance weighting and graph structure to be quantified. Empirical coverage also varies around the nominal target.

Additional temporal periods could show how score differences change over time. Paired comparisons could add uncertainty estimates for these differences while accounting for temporal dependence. Ablations with count-only weights and graph-free shrinkage could quantify the contributions of weighting and connectivity. Node-level coverage analysis

could show how coverage varies across locations. Another extension is to consider several nodes with limited histories, as might occur after a shared interruption in data collection. This would examine interval estimation when the target’s neighbors also have less data available for sharing.

Future work can explore privacy-preserving scale sharing, online calibration under distribution shift, learned or time-varying graphs, and joint prediction intervals across multiple locations. These directions could extend uncertainty estimation to networks with changing relationships and restricted data access.

VI. CONCLUSION

Prediction intervals express uncertainty alongside point forecasts, including at locations with limited recent data. We presented a method that combines variance-weighted graph

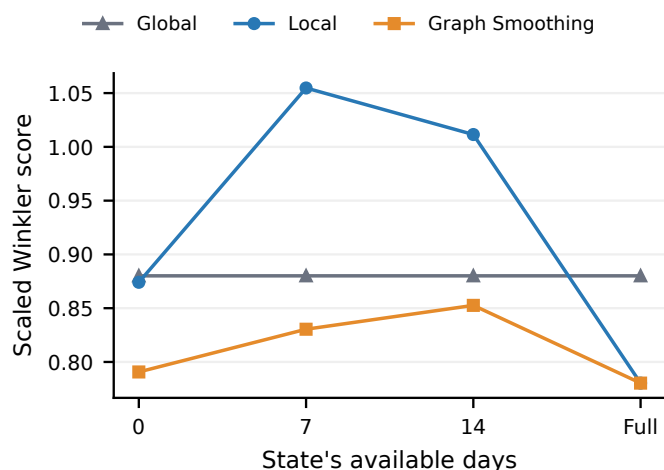


Fig. 5. Mean scaled Winkler score for seven-day-ahead hospital-admission forecasts. Local and Graph Smoothing overlap at Full.

smoothing of location-specific log-scales with calibration. In traffic forecasting and hospital admissions, the method lowers mean Winkler scores relative to separate location-specific scales when target histories are short, with gains over pooling that vary across datasets. This approach uses information from connected locations to support interval estimation when local data are scarce.

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